

FEDERATED LEARNING AND GROKKING: IS GROKKING PRESENT AND HOW DOES FEDERATION INTERACT WITH IT?

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ABSTRACT

An abstract abstract about abstract abstracts.

To do list (ordered by dependencies + most useful)

- Experimental findings figures
- Experimental findings writeup
- Appendices
- redo and expand conclusion and discussion
- Condense Intro
- Condense Related Work
- Experimental setup cleanup
- add summary/brief discussion of results and bullet points of the 3 key findings at end
- abstract
- clean Appendices
- clean up citations and bib file
- check ICLR formatting rule compliance

1 INTRODUCTION

Deep neural networks (Schmidhuber (2015); LeCun et al. (2015)) are typically expected to generalize as training progresses, with training loss decreasing and test performance improving in tandem. However, recent work on grokking (Power et al. (2022); Humayun et al. (2024a); Liu et al. (2023b); Lyu et al. (2024b)) has shown that this intuition can fail in striking ways. On small algorithmic datasets such as modular arithmetic, overparameterized models can first memorize the training set, reaching near-zero training error while test accuracy remains close to chance, and only much later undergo a sharp transition to strong generalization after prolonged optimization. This delayed generalization phenomenon has attracted significant interest because it provides a clean setting for studying the interplay between memorization, representation learning, optimization, and regularization in modern neural networks. Subsequent work has suggested that grokking is closely tied to structured representation learning, weight decay, and the evolution of weight norms.

At the same time, federated learning (FL) (Kairouz et al. (2021), correct CITATION NEEDED) has emerged as a standard framework for training models across decentralized clients. In canonical FL, clients perform steps of local optimization and a server periodically aggregates their updates, often through model averaging as in FedAvg, although many other approaches exist. Unlike centralized training, FL introduces multiple additional sources of complexity: data may be highly skewed and unevenly split across clients (G. et al. (2024)), only a subset of clients may participate in each round, and local compute steps can cause substantial client drift (Gao et al. (2022)) before resynchronization when the server aggregates changes. These effects have motivated a large literature on heterogeneous federated optimization, including methods such as FedProx (Li et al. (2020)) and SCAFFOLD (Karimireddy et al. (2021)) that explicitly aim to stabilize local training or correct drift under non-IID data. While this line of work has largely focused on convergence speed, robustness, and communication efficiency, much less is understood about how the distinctive optimization

(Zeng et al. (2025)) geometry of FL affects when and how generalization emerges (Caldarola et al. (2022b)).

These two threads naturally raise a compelling question: **does grokking still occur under federated optimization, and if so, how is its timing and mechanism altered by decentralization?** On the one hand, one may expect grokking to survive in federated settings whenever the global dataset still contains the underlying algorithmic structure. From this perspective, FL changes the optimization path but not the existence of the generalizing solution (Qu et al. (2022)). Moreover, periodic model averaging may itself act as a form of implicit regularization, biasing training toward simpler or more shared representations in a way loosely analogous to the role that weight decay plays in centralized grokking. On the other hand, there are equally plausible reasons for FL to disrupt delayed generalization. Client drift may repeatedly pull the model toward locally memorizing solutions; non-IID (Chung et al. (2026)) partitions may prevent individual clients from observing enough of the algebraic structure to build a global circuit; and repeated re-synchronization may interfere with the slow weight-norm and representation dynamics that prior grokking studies identify as central to the eventual transition.

This paper studies grokking through the lens of federated learning using a deliberately controlled setup. We focus on toy algorithmic and simpler classification tasks where grokking is already known to arise in centralized training, such as modular arithmetic and MNIST classification. This allows us to ensure that the effects are solely due to federated optimization, and not the choice of setup, since grokking is already highly sensitive to various factors such as training duration, dataset size, learning rate, and regularization. Federated learning introduces an additional layer of hyperparameters, including the number of clients, participation rate, local epochs, and aggregation algorithm that we ablate and consider. We first verify the base grokking result in the centralized setting, then partition the same task across multiple clients varying the factors mentioned above. We track both local and global training dynamics, and examine how key FL knobs such as the number of clients, local epochs, participation rate, and drift-correction method alter the onset of delayed generalization. We compare standard FedAvg against other algorithms such as SCAFFOLD and FedProx, among others, to test whether their different treatments of client drift affect whether grokking is preserved, delayed, accelerated, or suppressed. More broadly, this study aims to connect two separate literatures: grokking as a model of delayed generalization, and federated learning as a model of decentralized, heterogeneous optimization. Understanding their interaction may shed light on whether grokking is a fragile artifact of centralized training or a more robust phenomenon tied to the geometry of learning itself.

2 RELATED WORK

Federated Learning. Federated Learning (Beutel et al., 2022; McMahan et al., 2023) is a decentralized machine learning paradigm whereby multiple clients collaboratively train a shared global model while keeping their raw data localized. In canonical FedAvg algorithms, clients perform multiple optimization steps on their local data before parameters or updates are sent to and averaged by a central server (CITATION NEEDED). This local training dynamic and the heterogeneous data split across clients means that clients may have different local optimization objectives from the global objective, with this divergence called *client drift*. A number of federated optimization methods have since been developed to control or compensate for these deviations both at the local and server levels. Client-wise methods add constraining penalties for drift or use correct local updates for their drift (CITATIONS fedprox, scaffold), whilst server-side variants often apply adaptive optimization to alter the overall global optimization dynamics (CITATIONS (Reddi et al., 2021)).

Beyond convergence, works have investigated how federated optimization affects generalization properties of the learn model. In federated learning, heterogeneous training can often converge to sharper solutions, associated with poorer generalization (CITATIONS -(Caldarola et al., 2022a)), with flat local solutions not necessarily yielding a flat global solution (CITATION -(Lee & Yoon, 2024)). Other work has shown that distributed optimization can alter the implicit regularization present in centralized SGD, showing that the distributed optimization procedure itself can change the generalization properties of the resulting solution (Dandi et al., 2022). Theoretical research has also studied federated generalization. Work has derived stability-based generalization bounds for common federated regimes, showing them to depend on client heterogeneity and optimizer conver-

gence behaviour (check citation accurate - (Sun et al., 2024)), with local steps and server optimizer momentum also producing trade-offs between faster optimization and stability (Zeng et al., 2024). This theoretical work, however is largely considering final generalization performance or bounds, rather than the when and how emergence occurs during training. This work on grokking therefore provides a complementary setting to study how federated optimization alters the transition towards generalizing solutions.

Grokking. The term **grokking** (Power et al., 2022) was introduced and popularized by work from OpenAI, who observed that overparameterized networks trained on small algorithmic datasets can exhibit a long period of near perfect training accuracy with poor test accuracy which is followed by a sudden transition to strong generalization after many additional optimization steps. Subsequent work has shown that this behaviour extends beyond the original settings and is not restricted to specific architectures, tasks or optimizers (Liu et al., 2023a; Humayun et al., 2024b; Gromov, 2023a). A major line of work has connected this transition to the emergence of internal representations and regime phase transitions during training (check citations (Liu et al., 2022; Nanda et al., 2023; Gromov, 2023a)). Other work has focussed on why generalizing solutions emerge only after a prolonged delay, with a series of competing and complementary findings relating to weight norms, loss landscapes, neural network circuits, feature learning dynamics and implicit bias (check citations, need more correcting probably - Liu et al. (2023a); Varma et al. (2023); Kumar et al. (2024); Lyu et al. (2024a)). Although these accounts differ in the precise mechanism, they all broadly suggest the grokking depends on the evolution of the optimization trajectory after memorization, which eventually favours a more structured generalizing solution. Existing work has, however, overwhelmingly studies these dynamics under centralized training and optimization, leaving open how grokking is affected, or whether it is even present, when learning is distributed across multiple local trajectories.

3 EXPERIMENTS

To explore the effect of learning federation on grokking, we split key factors into two groups: data fragmentation and learning fragmentation, where data fragmentation refers to how the data is split up across clients (e.g., number of samples per client, distribution, spread) and learning fragmentation refers to the optimization aspect where clients learn separately and the resulting updates are combined.

3.1 EXPERIMENTAL SETUP

To explore how different elements of federated learning affect grokking, we use and adapt five different experimental setups from known grokking works (CITATIONS CHECK nanda citation, others? Gromov (2023b); Liu et al. (2023a)). These consist of both multi-layer perceptron and transformer-based models, with differing optimizers, datasets, and architectures in order to provide generalisable findings across federated settings. To summarise we use: **SETUPS USED**.

See full specific details about each setup in **LINK to APPENDICES**. All experimental runs are repeated and tested across three different seeds to prevent any outlier seeds from skewing results. To perform federated training, we use the Flower library (Beutel et al. (2022)) to simulate training across multiple devices.

3.2 DATA-WISE FRAGMENTATION

In federated learning, data is not globally known or uniformly distributed. Instead, different clients have different subsets of data, with such subsets often biased towards some classes or characteristic. This section explores how the effect of splitting data in this manner, and recombining the corresponding weight updates, effects the grokking phenomenon.

3.2.1 NUMBER OF CLIENTS

We first examine whether grokking persists under federated learning as the number of clients K increases under *iid* partitioning. Intuitively, a larger number of clients K reduces the amount of

Figure 1: **Federated slowdown of grokking relative to centralized training.** We plot the ratio $T_{\text{grok}}^{\text{FL}}/T_{\text{grok}}^{\text{cent}}$ as a function of the number of clients K , under different fractions of training data α for IID data. A ratio of 1 indicates no slowdown relative to centralized training. In most settings, increasing the number of clients delays the grokking transition, with the effect becoming especially pronounced for smaller α and large K . Error bars indicate variation across runs.



data available to each client and so increases the mismatch between local training trajectories and the global objective, which may delay or disrupt the transition from memorization to generalization.

As shown in FIGURE X, federated models still grok under IID partitioning, even in regimes where centralized models trained on only $1/K$ of the data fail to do so. This indicates that aggregation of learning across clients can retain the global structure needed for generalization. However, grokking in federated settings is often, but interestingly not always, slower than centralized grokking, and the slowdown increases with K . For small K , the slowdown ratio remains close to 1, while for larger K it rises substantially, especially at smaller α . In the most extreme setting, the slowdown exceeds $2\times$.

These results suggest that grokking is preserved but systematically delayed in federated settings. As the number of clients grows, each client trains on a smaller local subset and drifts further toward local memorization before aggregation, making the global transition to a generalizing solution slower and less stable.

3.2.2 DATA HETEROGENEITY

We next study whether *non-IID* partitioning changes the emergence of grokking. Since real federated learning deployments are typically heterogeneous, an important question is whether delayed generalization is merely slowed down or fundamentally disrupted when clients observe skewed local data.

For **data heterogeneity experiment**, we vary the Dirichlet concentration parameter $\alpha_{\text{dir}} \in \{0.01, 0.1, 0.5, 1.0, 1000\}$ at fixed $K = 10$ and $E = 5$, across training fractions $\alpha \in \{0.25, 0.30, 0.35, 0.50\}$. Smaller α_{dir} corresponds to stronger heterogeneity, while large α_{dir} approaches IID partitioning. As shown in Figure 2, heterogeneity has only a mild effect in most regimes: T_{grok} is largely stable once $\alpha_{\text{dir}} \geq 0.1$. However, under extreme heterogeneity ($\alpha_{\text{dir}} =$

Figure 2: **Effect of data heterogeneity on federated grokking.** We plot the grokking time T_{grok} versus the Dirichlet concentration parameter α_{dir} for $K = 10$ clients, under different training fractions α . Smaller α_{dir} corresponds to stronger non-IID partitioning, while large α_{dir} approaches IID. Across most settings, grokking time is relatively stable under moderate heterogeneity, but extreme heterogeneity ($\alpha_{\text{dir}} = 0.01$) noticeably delays the grokking transition, especially at smaller training fractions. Error bars indicate variation across runs.

figure/main/exp3a_t_grok_vs_dir_alpha.png

0.01), grokking is noticeably delayed, with the slowdown most pronounced at smaller training fractions. And we compare IID partitioning against two structured non-IID splits: an *operand split*, where clients are partitioned by equations of the form $a + b = n$, and a *target split*, where clients are partitioned by target classes $(a + b) \bmod p = n$. We find that these structured partitions produce similar T_{grok} to IID, indicating that the precise partition structure has little effect in our setting. Instead, the dominant factor remains the training fraction α .

Overall, non-IID partitioning does not prevent grokking, but sufficiently strong heterogeneity can delay its onset. This suggests that delayed generalization in federated learning is robust to moderate heterogeneity, while extreme client skew mainly acts as an additional source of optimization slowdown.

3.3 LEARNING FRAGMENTATION

Federated learning also splits learning across different client models, with each learning updates on their respective data subsets. This updates accumulate for a set number of local epochs, before being sent back to the central node to combine and send the new model to use as a base for the next set of local epochs. during each set of local epochs, models drift and diverge from each other. we aim to explore how this effects the grokking phenomenon.

3.3.1 LOCAL EPOCHS

We next study how the number of local epochs E affects grokking in federated learning. Since larger E allows each client to optimize further before aggregation, it increases client drift and may move local models away from the global trajectory needed for delayed generalization.

Figure 3: Effect of local epochs on federated grokking. We plot grokking time T_{grok} versus the number of local epochs E for $K = 10$, under different training fractions α and both IID and non-IID ($\alpha_{\text{dir}} = 0.1$) partitions. Increasing E generally delays grokking, with a stronger effect at smaller α and under non-IID data; in some low-data settings, the model fails to grok within the training budget. Error bars denote variation across runs.



We sweep $E \in \{1, 5, 10, 25, 50\}$ for $K = 10$ across training fractions $\alpha \in \{0.25, 0.30, 0.50\}$ under both IID and non-IID partitioning. As shown in Figure 3, local epochs are a key factor: small values ($E = 1, 5$) preserve grokking, while larger values increasingly delay the transition. This effect is strongest at smaller α and under non-IID data. In low-data boundary regimes, large E can prevent grokking entirely within the training budget.

This trend is consistent with the role of client drift in federated optimization. Empirically, the drift magnitude $\|w_k - w_{\text{global}}\|$ grows with E and is strongly correlated with delayed or failed grokking ($r = 0.85$), suggesting that excessive local training disrupts the slow transition from memorization to generalization.

3.3.2 PARTIAL PARTICIPATION

We also test whether reduced client participation changes grokking dynamics. In practical federated learning, only a fraction of clients may participate in each round, which introduces additional sampling noise and reduces the amount of data contributing to each aggregation step.

We sweep the participation fraction $f \in \{0.2, 0.4, 0.6, 1.0\}$ for $K = 10$ and $E = 5$ under both IID and non-IID partitioning. As shown in Figure 4, varying f has little effect on T_{grok} across all training fractions α . Even at 20% participation, grokking occurs at nearly the same time as full participation.

These results suggest that, unlike local epochs, partial participation is not a major factor in delayed generalization in our setting. Grokking appears robust to moderate client subsampling, indicating that client drift rather than participation noise is the dominant optimization bottleneck.



Figure 4: **Effect of client participation on federated grokking.** We plot grokking time T_{grok} versus the client participation fraction f for $K = 10$ and $E = 5$, under different training fractions α and both IID and non-IID ($\alpha_{\text{dir}} = 0.1$) partitions. Across all settings, varying f has little effect on the timing of grokking. Error bars denote variation across runs.

3.4 ANALYSIS

To understand why federated grokking is delayed or fails, we perform a post-hoc analysis over all runs from all experiments, focusing on three mechanistic indicators: client drift, parameter norm structure, and Fourier concentration.

First, **client drift is the main federated perturbation**. As shown in Figure 5, larger mean drift is consistently associated with larger T_{grok} , while factors that induce little change in drift (e.g., participation fraction) have little effect on delayed generalization. This matches the earlier empirical trends: increasing K or E slows grokking primarily because it increases the mismatch between local client trajectories and the global solution path.

However, drift alone does not explain the mechanism. Figure 6 shows that larger drift does *not* simply prevent weight growth. Instead, it changes *how* the model grows. In particular, stronger drift is associated with a persistently smaller layer-norm ratio $\|W_2\|/\|W_1\|$, suggesting that client updates over-amplify early-layer growth while failing to develop the higher-layer structure associated with grokking. Consistent with this, Figure 7 shows that memorization time is weakly informative: most runs fit the training set early, but their grokking times vary widely. Thus, the key bottleneck is not memorization itself, but the subsequent emergence of a structured generalizing solution.

The strongest indicator of this structure is **Fourier concentration**, measured by IPR. Across runs, faster grokking is associated with earlier and larger IPR growth, while delayed or failed grokking corresponds to suppressed IPR trajectories (Figure 6). Taken together, these results suggest the following mechanism: federated client drift does not stop norm growth, but *misdirects* it, disrupting the parameter scaling needed for the structured Fourier solution to emerge. In extreme cases, this leads not only to slower grokking, but to effectively intractable delays.

This interpretation is further supported by the mitigation experiments. Figure 8 shows that FedAdam substantially outperforms FedAvg and FedProx in hard grokking regimes, while explicit weight decay fails to recover grokking. Figure 9 explains why: FedAdam achieves lower sustained drift, rapidly increases $\|W_2\|/\|W_1\|$, and exhibits early IPR growth, indicating that it steers training to-

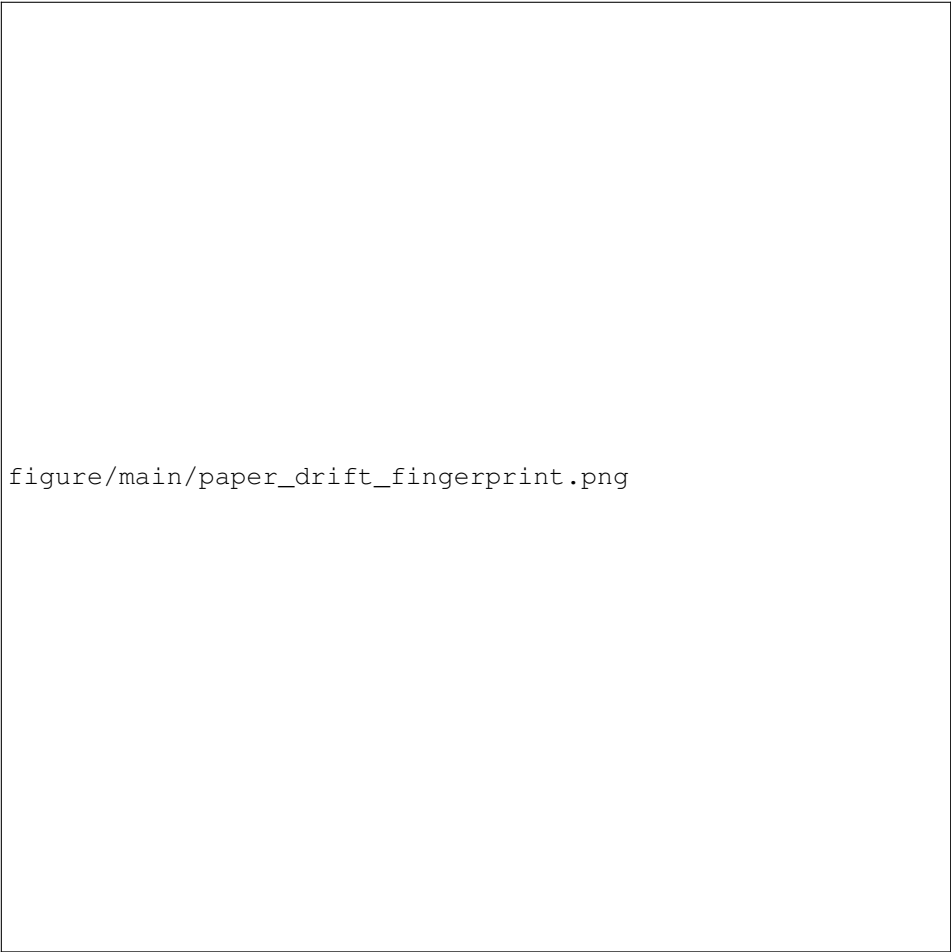
Figure 5: **Client drift strongly correlates with delayed grokking.** We plot grokking time T_{grok} against mean client drift under four sweeps: (a) local epochs E , (b) number of clients K , (c) Dirichlet heterogeneity α_{dir} , and (d) participation fraction f . Across settings, larger drift is consistently associated with later grokking, while sweeps that induce little change in drift (especially participation) have little effect on T_{grok} .

figure/main/drift_vs_grokking.png

ward the correct structured solution rather than merely reducing optimization noise. Overall, our analysis suggests that delayed generalization in federated learning is governed by whether optimization preserves the internal norm and Fourier dynamics required for grokking.

4 CONCLUSION & DISCUSSION

In this paper, we studied whether grokking persists under federated learning and what controls its delay. On modular arithmetic, we find that federated models can still grok, but the transition is generally slower than in centralized training. The main factors are the number of clients and local epochs, while moderate partial participation has little effect and heterogeneity mainly matters in extreme non-IID settings. Our analysis points to *client drift* as the key federated perturbation. Larger drift is associated with later grokking, not because it prevents memorization, but because it disrupts the parameter scaling and Fourier structure needed for generalization. Mitigation results further show that optimization matters: FedAdam can substantially accelerate grokking in hard regimes. Our study is limited to algorithmic toy tasks, mostly $a + b \bmod 97$, although experiments show that our findings generalize to other modular arithmetic tasks, and a two-layer MLP setup. Future work should test whether the same mechanisms hold for larger models, other tasks, and more realistic federated benchmarks.



figure/main/paper_drift_fingerprint.png

Figure 6: **Layer-norm ratio under different local learning regimes.** We plot the evolution of the layer-norm ratio $\|W_2\|/\|W_1\|$ over training for centralized learning and federated training with different numbers of local epochs E . Larger E yields a persistently smaller ratio, indicating that stronger client drift alters the internal parameter scaling associated with grokking.

Figure 7: **Grokking time is not explained by memorization time alone.** We plot grokking time T_{grok} against memorization time T_{mem} across experiments. Most runs achieve memorization early, while the onset of grokking varies widely, indicating that delayed generalization is largely decoupled from the time required to fit the training set.



5 USE OF AI

We used AI tools to assist with refining the language and improving the clarity of our writing. We also used them in to assist with code development.

6 NOTES

contributions summarised: 1. grokking in federating learning setting. 2. we identify FL specific learning dynamics for grokking, via FL specific hyper parameters: number of clients, participation ratio etc. 3. we examined FL techniques to mitigate client drift and see how that affects grokking: larger drift means later and grokking, because it disrupts existing network structures needed for generalisation. 4. we perform mech interp analyses why client results in later grokking.

extra runs needed: 1. mech interp stuff needed, model weights (do last) 2. partial participation work 3. local epochs not looked into 4. iid ness also not looked into.

- change/play with E and K for figures 3, 4

A - original work from miniproject A' - A but with adamw optimiser (can completely ignore/dont do for all future runs) B - nanda transformer C- transformer on different dataset D - quadratic activation MLP on different dataset E - MNIST omnigrok setup

Figure 8: **Alternative federated optimizers in hard grokking regimes.** We plot test accuracy over training for three challenging settings (H1–H3) under FedProx, FedAdam, and FedAvg with explicit weight decay. FedProx partially improves grokking in some cases, FedAdam substantially accelerates the transition to generalization, while FedAvg with additional weight decay does not recover grokking in these settings.

figure/main/exp5_training_curves_no_fedavg.png

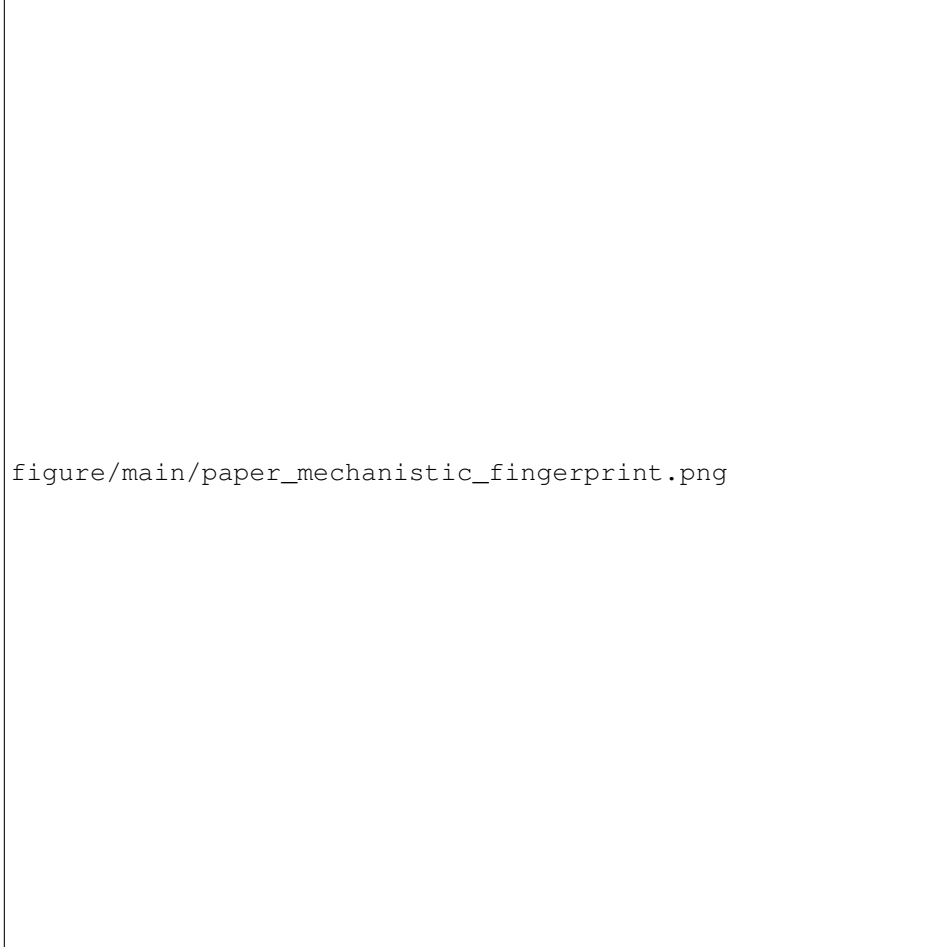


Figure 9: **Mechanistic fingerprints of alternative federated optimizers.** We compare FedAvg, FedProx, FedAdam, and FedAvg with explicit weight decay on a hard grokking setting, showing test accuracy, client drift, layer-norm ratio $\|W_2\|/\|W_1\|$, and IPR over training. Different optimizers induce distinct optimization trajectories, with faster grokking associated with lower sustained drift and earlier growth of internal structure.

optimiser persist = false this only matters for not pure GD

<https://claude.ai/code/artifact/b8668f34-204f-4ec9-bec0-4d38c1a7f5fc>

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